

HANYU YANG

CRM DATA ANALYTICS / SOFTWARE SYSTEMS / APPLIED AI

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PROFILE

Software engineering graduate and incoming postgraduate student in AI, Ethics and Society at the University of Hong Kong. Experience spans CRM data analytics in financial services, AI workflow development, full-stack products, multimodal research, and distributed data systems. I work best where technical evidence, business context, and clear delivery meet.

01 / EXPERIENCE

Professional Experience

HSBC - Wealth and Personal Banking

CRM Data Analyst Intern

Mar. 2026 - Jul. 2026

Guangzhou, China

- Integrated core-banking, credit-card, and mobile-channel customer data with SQL and Python on GCP BigQuery and Data Science Workbench, supporting a unified customer view and precise WPB profiling.
- Performed data cleaning, exploratory analysis, and predictive modeling in Python to support customer segmentation and evidence-led marketing decisions.
- Contributed to precision-marketing strategy and campaign deployment through SmartCRM and PEGA, supporting omnichannel customer lifecycle management.
- Supported omnichannel customer-communications delivery by coordinating handover requirements, UAT inputs, execution readiness, channel schedules, and stakeholder sign-offs across in-app, eDM, and marketing-push journeys.

TOOLS SQL / Python / GCP BigQuery / Data Science Workbench / SmartCRM / PEGA / UAT

Guangzhou Xin'an Data Co., Ltd.

AI Development Intern - Digital Innovation Department

Jul. 2025 - Aug. 2025

Guangzhou, China

- Designed and deployed a full-stack AI prompt-optimization platform with authentication, community libraries, and knowledge-sharing capabilities for more than 100 internal users.
- Integrated Model Context Protocol with Dify to automate AI-driven reporting workflows.
- Developed RESTful APIs for third-party integration and enterprise data exchange.

TOOLS MCP / Dify / REST API / Full-stack systems / AI workflows

02 / EDUCATION

Education

Master of Arts in AI, Ethics and Society

The University of Hong Kong - Hong Kong SAR

Sept. 2026 - Aug. 2027
(Expected)

Incoming postgraduate study connecting AI systems with their ethical and societal contexts.

Bachelor of Science in Software Engineering

Guangdong University of Foreign Studies - School of Information Science and Technology

Sept. 2022 - Jun. 2026

GPA 88/100 - Top 20%. Selected modules: Text Information Processing (99), Network Programming (96), Natural Language Processing (96), Software Architecture (96), Big Data Processing Technology (94), Further Mathematics (91), and Web Design (90).

Research Experience

Guangdong University of Foreign Studies

Oct. 2024 - Dec. 2024

Research Contributor - Fake News Detection through Multimodal Attention

Guangzhou, China

- Proposed a semantic-conflict recognition and multimodal-attention framework for image-text inconsistency.
- Added KL-divergence and MSE modules to CLIP alignment for quantifying cross-modal semantic discrepancies.
- Combined BERT, ResNet-50, and cross-modal self-attention, reaching 88.9% accuracy and improving on the CLIP baseline by 2.8%.

TOOLS BERT / ResNet-50 / CLIP / Multimodal attention

University of Cambridge

Aug. 2024

Research Leader - Skin Cancer Classification with CNN and Attention

Cambridge, UK

- Led development of a lightweight enhanced-AlexNet CNN for benign-versus-malignant dermoscopic lesion classification.
- Integrated spatial attention to focus the model on diagnostically relevant regions.
- Improved classification accuracy by 15% through preprocessing optimization and background-noise suppression.

TOOLS CNN / AlexNet / Spatial attention / Medical imaging

Guangdong University of Foreign Studies

Oct. 2023 - Nov. 2023

Algorithm Designer - Autonomous Parking Detection

Guangzhou, China

- Redesigned YOLOv5s with a GhostNet backbone and CBAM attention for parking-slot detection under occlusion and variable lighting.
- Trained in PyTorch with RMSProp and Mosaic augmentation for 500 epochs on RTX 3090.
- Reached 97.1% AP and 93.7% AR while reducing inference latency by 20% against the YOLOv5s baseline.

TOOLS PyTorch / YOLOv5s / GhostNet / CBAM

Selected Engineering Projects

International Chinese Education Platform - Front-End Developer

Apr. 2025 - Jul. 2025

Vue 3 / Composition API / Real-time UX

Built a responsive student portal with real-time class management and messaging for more than 500 international students. Added lazy loading and virtual scrolling for smooth interaction under real-time data flow.

YouTube English Comment Sentiment Analysis - Project Leader

May. 2025 - Jun. 2025

NLTK / TF-IDF / Word2Vec / LinearSVC / XGBoost / Streamlit

Collected more than 50,000 comments across five topics; built and tuned classical-ML feature pipelines. Reported 90.5% validation accuracy and 0.89 F1; deployed a sub-two-second Streamlit interface for 100+ internal users.

Global Economic Data Analysis - Data Engineer

Nov. 2024 - Dec. 2024

Hadoop / Hive / Spark / Flink / Kafka

Processed 8,519 records with 27 features on a three-node Hadoop cluster. Built batch and streaming pipelines across 150+ countries; reduced complex batch-query response time by 40% and improved time-series queries by 15%.

ChatGDUFS Campus Q&A - NLP Developer

Nov. 2024 - Dec. 2024

LangChain / RAG / Chroma / Qwen-7B / LoRA / Flask

Built a RAG pipeline with 2,000+ document embeddings and a sub-two-second Flask query API. Fine-tuned Qwen-7B with LoRA on 5,000 campus Q&A pairs, with reported domain accuracy improving from 72% to 86%.

Technical Toolkit

PROGRAMMING	FRAMEWORKS & SYSTEMS	LANGUAGES
C++ / Java / Python / SQL / Vue 3 / React / JavaScript / Git	TensorFlow / PyTorch / MySQL / Docker / Redis / Hadoop / Spark / Flink / Kubernetes	English - IELTS 7 Mandarin - Native Cantonese - Fluent

Awards & Recognition

01	China International College Students' Innovation Competition - Outstanding Award, Jul. 2025.
02	14th Challenge Cup Guangdong College Students Entrepreneurship Competition - Bronze Award, Jul. 2024.
03	National College Students English Vocabulary Ability Competition - Third Prize (National), Jun. 2024.
04	Guangdong University of Foreign Studies - Third-Class Scholarship for Outstanding Students, Oct. 2023.
05	Guangdong Youth Science and Technology Innovation Competition - Silver Award, Oct. 2019.

Publication

Yang, H., Bai, J., He, C., Wang, Y., & Liang, W. (2024). *Synthesis, characterization, and testing of rare earth-doped GdBO3 luminescent materials. Guangdong Chemical Industry*, 51(10), 50-54.

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